

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

ON

EVENTSNAP

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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CONTENTS

CONTENTS

S.No.	Title	Page No.
1.	Declaration	3
2.	Acknowledgement	4
3.	Abstract	5
4.	Introduction	6
5.	Methodology	7-8
6.	Tools and Technologies	9-10
7.	Implementation	11-12
8.	Major Findings / Outcomes / Results	13
9.	Conclusion and Future Scope	14
10.	References	15

DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**EventSnap**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Ajay Kumar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

Abhishek Kachhap(2210991165)

Date: 08.05.2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Place: Rajpura

Date: 08.05.2026

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ABSTRACT

EventSnap – Where Every Face Finds Its Photo is an AI-powered web application developed to simplify the process of finding personal photographs from large collections of event images. In events such as weddings, college fests, parties, seminars, and social gatherings, thousands of photographs are captured, making it difficult for individuals to manually search for photos in which they are present.

The main objective of this project is to automate the photo discovery process using face recognition technology. The platform allows event organizers to upload event photographs, while users can upload a selfie or personal image to search for matching photos. The system detects and compares facial features using computer vision and machine learning techniques and then displays all matched photographs to the user.

The project is developed using the MERN Stack, which includes MongoDB, Express.js, React.js, and Node.js. React.js is used for building an interactive and responsive frontend user interface, while Node.js and Express.js handle backend services and API development. MongoDB is used for storing user data, event information, and image metadata securely. Face recognition functionality is implemented using face-api.js and TensorFlow.js for detecting and matching facial embeddings.

JWT authentication is implemented to provide secure login and role-based access control. Cloudinary is used for secure image storage and image management. RESTful APIs are used for communication between frontend and backend modules.

The platform reduces manual effort, improves accessibility of event memories, and provides a smart, secure, and scalable solution for event photo management. EventSnap demonstrates the practical implementation of Artificial Intelligence and Full Stack Development in solving real-world problems related to event photography and image retrieval.

INTRODUCTION

EventSnap – Where Every Face Finds Its Photo is an AI-based event photo retrieval system designed to help users quickly identify and access photographs in which they appear. In modern social events such as weddings, festivals, conferences, farewell parties, and college events, photographers capture hundreds or thousands of images. Searching for personal photographs manually becomes time-consuming and inefficient.

Traditional methods of sharing event photos usually involve uploading complete albums to cloud drives or social media platforms. Users must manually scroll through large image collections to locate their own photographs. This process becomes inconvenient, especially when the number of photographs is extremely high.

To solve this problem, EventSnap uses Artificial Intelligence and Face Recognition Technology to automate the image searching process. Users only need to upload a selfie, and the system intelligently compares the uploaded face with faces present in the event image database. The platform then retrieves all matched photographs instantly.

The project focuses on:

- Reducing manual effort in photo searching
- Improving user convenience
- Providing secure image management
- Enhancing event memory accessibility
- Automating image retrieval using AI

The system also provides event-based access control using event codes, ensuring privacy and controlled access to photographs.

EventSnap combines Full Stack Web Development with AI-powered facial recognition to create a practical and innovative real-world application.

METHODOLOGY

The development of EventSnap followed a structured software development methodology consisting of requirement analysis, system design, implementation, testing, and deployment. Agile methodology was adopted to ensure continuous improvement and flexibility during development.

2.1 Requirement Analysis

The first phase involved identifying the major requirements of users and event organizers.

Functional Requirements

- User Registration and Login
- Event Creation
- Event Code Generation
- Selfie Upload
- Event Image Upload
- Face Detection
- Face Matching
- Photo Retrieval
- Secure Image Storage
- Gallery View

Non-Functional Requirements

- Secure Authentication
- Fast Search Performance
- Responsive UI
- Scalable Architecture
- Data Privacy
- Cloud Storage Support

2.2 System Design

The system architecture follows a three-tier architecture consisting of:

Frontend Layer

Responsible for user interaction and UI rendering.

Backend Layer

Handles APIs, authentication, image processing, and business logic.

Database Layer

Stores user data, image metadata, and event details.

2.3 Face Recognition Workflow

The face recognition module works in the following steps:

1. User uploads selfie
2. System detects facial landmarks
3. Facial embeddings are generated
4. Event images are scanned
5. Facial embeddings are compared
6. Matching threshold is applied
7. Matched images are displayed

2.4 Development Approach

Agile development methodology was used:

- Requirement Planning
- Module Development
- Incremental Testing
- UI Improvements
- Deployment Preparation

Tools and Technologies

Technology	Purpose
React.js	Frontend User Interface
FastAPI (Python)	Backend API Development
MongoDB Atlas	Database Management
JWT Authentication	Secure Login & Authorization
Tailwind CSS	Styling & Responsive Design
face-api.js	Face Detection & Recognition
TensorFlow.js	Machine Learning Processing
Cloudinary	Cloud Image Storage
Axios	API Communication
Postman	API Testing
Git & GitHub	Version Control
VS Code	Development Environment

React.js

React.js is used for developing responsive and component-based frontend interfaces. It improves maintainability and provides a smooth user experience.

Node.js and Express.js

Node.js provides backend runtime support, while Express.js simplifies server-side routing and API development.

Technology

Purpose

MongoDB Atlas

MongoDB Atlas is used as a cloud-based NoSQL database for storing:

- User details
- Event information
- Image metadata
- Face matching data

FastAPI (Python)

FastAPI is used for backend development and REST API creation. It provides high performance, asynchronous request handling, automatic API documentation, and efficient integration with AI and machine learning workflows. FastAPI handles authentication, image processing requests, database communication, and face matching APIs within the system.

Implementation

User Authentication Module

The system provides secure authentication using JWT tokens.

Features

- User Registration
- Login/Logout
- Protected Routes
- Role-Based Access

Passwords are encrypted before storage.

Event Management Module

Event organizers can:

- Create events
- Generate unique event codes
- Upload group photos
- Manage uploaded galleries

Face Recognition Module

This is the core module of the project.

Workflow

- User uploads selfie
- System detects face
- Face embeddings are generated
- Event images are scanned
- Similarity comparison is performed
- Matching photos are returned

Face Matching Algorithm

The Euclidean distance method is used for comparing facial embeddings.

If similarity threshold is satisfied:

- Image is considered matched

Image Upload and Storage Module

Images are uploaded using Cloudinary integration.

Features

- Secure cloud storage
- Optimized image delivery
- Fast retrieval
- Scalable image handling

Gallery Module

Users can:

- View matched images
- Download photos
- Access personal event memories

Admin Module

The admin dashboard provides:

- User Management
- Event Monitoring
- Image Moderation
- Database Monitoring

API Integration

REST APIs were developed for:

- Authentication
- Event Management
- Image Upload
- Face Matching
- Gallery Retrieval

Axios is used for frontend-backend communication.

Major Findings / Outcomes / Results

The EventSnap system successfully automates event photo searching using Artificial Intelligence and Face Recognition Technology.

Major Findings

Faster Photo Retrieval

Users can instantly locate their photographs without manually searching through large image collections.

Improved User Experience

The platform provides a simple and responsive interface for quick image discovery.

Reduced Manual Effort

Automated face matching eliminates the need for manual searching.

Secure Image Management

JWT authentication and cloud storage improve security and privacy.

Scalable Architecture

The MERN Stack architecture supports scalability for handling large event datasets.

Real-World Practicality

The project demonstrates practical implementation of AI in event photography management.

Conclusion and Future Scope

EventSnap successfully demonstrates the integration of Artificial Intelligence with Full Stack Web Development to solve real-world problems related to event photography and image retrieval.

The project simplifies the process of finding personal photographs in large event albums by using AI-powered face recognition technology. The system improves accessibility, reduces manual effort, and provides users with a secure and efficient platform for retrieving event memories.

The MERN Stack architecture ensures scalability, maintainability, and responsiveness. The implementation of face-api.js and TensorFlow.js demonstrates the practical application of machine learning in web applications.

The project also highlights the importance of secure authentication, cloud storage, and efficient API integration in modern web development.

The project can be enhanced further with advanced features such as:

Mobile Application

Developing Android and iOS applications for better accessibility.

Multi-Face Search

Allowing users to search for multiple people simultaneously.

Social Media Integration

Allowing direct sharing of matched photos to social media platforms.

Cloud Scalability

Using advanced cloud infrastructure for handling large-scale events.

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